Customer Service Requests Analysis

Create Sections

1. Import Libraries

2. Load Dataset

3. Understand Dataset

4. Data Cleaning

5. Exploratory Data Analysis

6. Complaint Analysis

7. Statistical Testing

8. Observations

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

%matplotlib inline
```

In [2]:
```
df = pd.read_csv("311_Service_Requests_from_2010_to_Present.csv")
df.head()
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_9444\1741486433.py:1: DtypeWarning: Columns
(0: Ferry Direction, 1: Ferry Terminal Name) have mixed types. Specify dtype option
on import or set low_memory=False.
  df = pd.read_csv("311_Service_Requests_from_2010_to_Present.csv")
```

Out[2]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Lo
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Stre
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access	Stre
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access	Stre
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Stre
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Stre

5 rows × 53 columns

In [3]:
```
df.columns
```

Out[3]:
```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='str')
```

In [4]:
```
df.shape
```

Out[4]:
```
(364558, 53)
```

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
Unique Key                           0
Created Date                         0
Closed Date                       2381
Agency                               0
Agency Name                          0
Complaint Type                       0
Descriptor                        6501
Location Type                      133
Incident Zip                      2998
Incident Address                 51699
Street Name                      51699
Cross Street 1                   57188
Cross Street 2                   57805
Intersection Street 1           313438
Intersection Street 2           314046
Address Type                      3252
City                              2997
Landmark                        364183
Facility Type                     2389
Status                               0
Due Date                             3
Resolution Description               0
Resolution Action Updated Date    2402
Community Board                      0
Borough                              0
X Coordinate (State Plane)        4030
Y Coordinate (State Plane)        4030
Park Facility Name                   0
Park Borough                         0
School Name                          0
School Number                        0
School Region                        1
School Code                          1
School Phone Number                  0
School Address                       0
School City                          0
School State                         0
School Zip                           1
School Not Found                     0
School or Citywide Complaint    364558
Vehicle Type                    364558
Taxi Company Borough            364558
Taxi Pick Up Location           364558
Bridge Highway Name             364261
Bridge Highway Direction        364261
Road Ramp                       364296
Bridge Highway Segment          364296
Garage Lot Name                 364558
Ferry Direction                 364557
Ferry Terminal Name             364556
Latitude                          4030
Longitude                         4030
Location                          4030
dtype: int64
```

In [6]:
```
df.columns = df.columns.str.strip()
df.columns
```

Out[6]:
```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='str')
```

In [7]:
```
df = df[df['Closed Date'].notnull()]
df.shape
```

Out[7]:
```
(362177, 53)
```

In [8]:
```
df['Created Date'] = pd.to_datetime(df['Created Date'], errors='coerce')
df['Closed Date'] = pd.to_datetime(df['Closed Date'], errors='coerce')
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_9444\12745884.py:1: UserWarning: Could not
infer format, so each element will be parsed individually, falling back to `dateutil
`. To ensure parsing is consistent and as-expected, please specify a format.
  df['Created Date'] = pd.to_datetime(df['Created Date'], errors='coerce')
C:\Users\HP\AppData\Local\Temp\ipykernel_9444\12745884.py:2: UserWarning: Could not
infer format, so each element will be parsed individually, falling back to `dateutil
`. To ensure parsing is consistent and as-expected, please specify a format.
  df['Closed Date'] = pd.to_datetime(df['Closed Date'], errors='coerce')
```

In [9]:
```
df[['Created Date','Closed Date']].isnull().sum()
```

Out[9]:
```
Created Date    0
Closed Date     0
dtype: int64
```

In [10]:
```
df = df[df['Closed Date'] >= df['Created Date']]
df.shape
```

Out[10]:
```
(362177, 53)
```

In [11]:
```
df['Request_Closing_Time'] = (
    df['Closed Date'] - df['Created Date']
).dt.total_seconds()
```

In [12]:
```
df['Request_Closing_Time'].describe()
```

Out[12]:
```
count    3.621770e+05
mean     1.511330e+04
std      2.110255e+04
min      6.100000e+01
25%      4.533000e+03
50%      9.616000e+03
75%      1.887800e+04
max      2.134342e+06
Name: Request_Closing_Time, dtype: float64
```

In [13]:
```
df['City'] = df['City'].fillna("Unknown City")
```

In [14]:
```
df['City'].isnull().sum()
```

Out[14]:
```
np.int64(0)
```

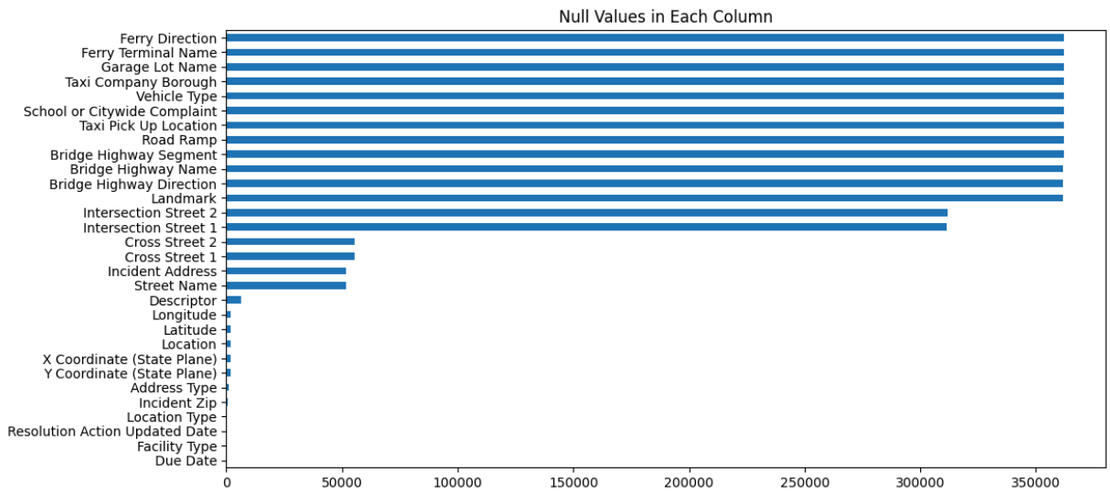
In [15]:
```
df = df.drop(columns=[
    'School Name','School Number','School Region',
    'School Code','School Phone Number',
    'School Address','School City','School State',
```

```
    'School Zip','School Not Found'
], errors='ignore')
```

NULL values graph

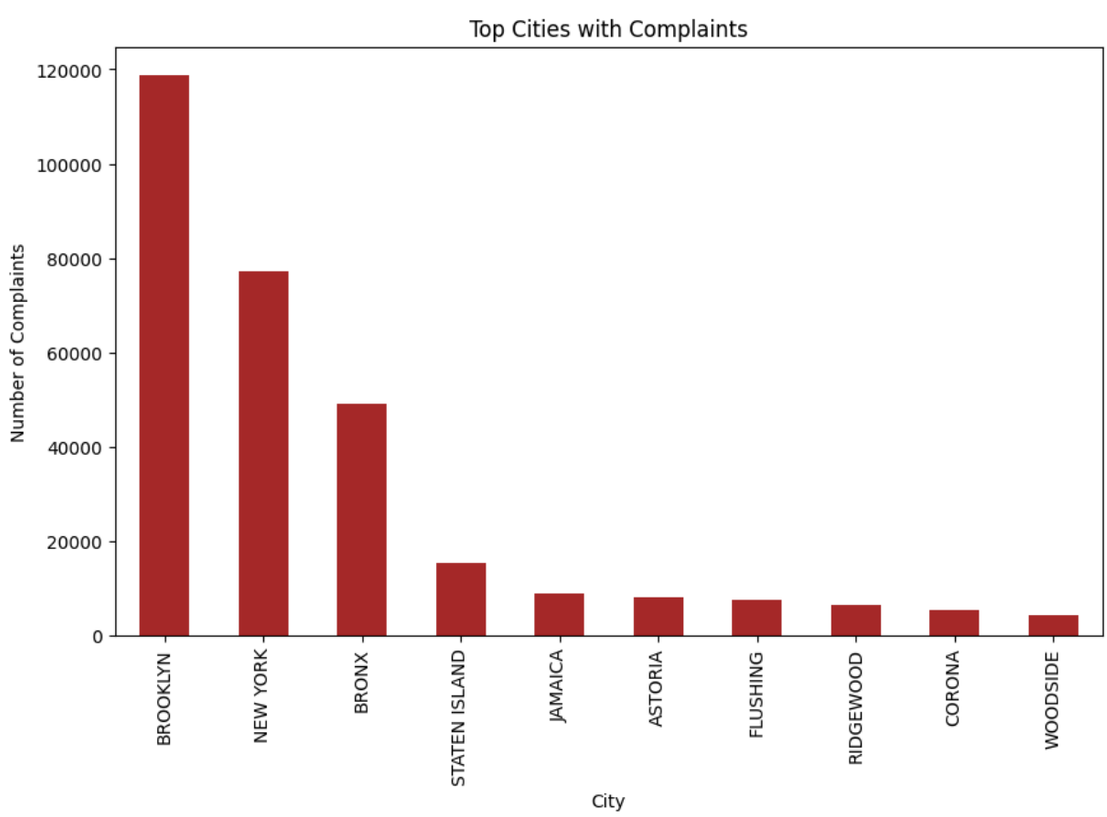
In [16]:
```
null_values = df.isnull().sum()
null_values = null_values[null_values > 0]

plt.figure(figsize=(12,6))
null_values.sort_values().plot(kind='barh')
plt.title("Null Values in Each Column")
plt.show()
```



Complaints in Each City

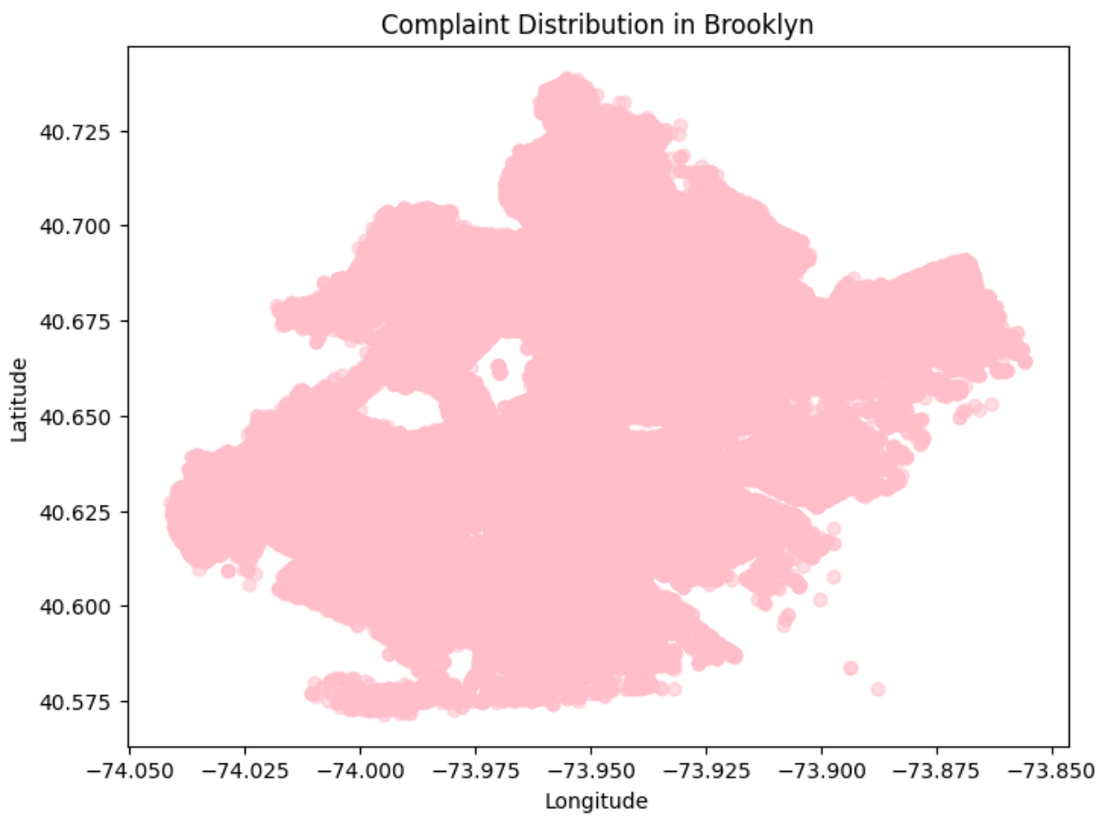
In [17]:
```
plt.figure(figsize=(10,6))
df['City'].value_counts().head(10).plot(kind='bar', color = 'brown')
plt.title("Top Cities with Complaints")
plt.xlabel("City")
plt.ylabel("Number of Complaints")
plt.show()
```



Scatter Plot (Brooklyn)

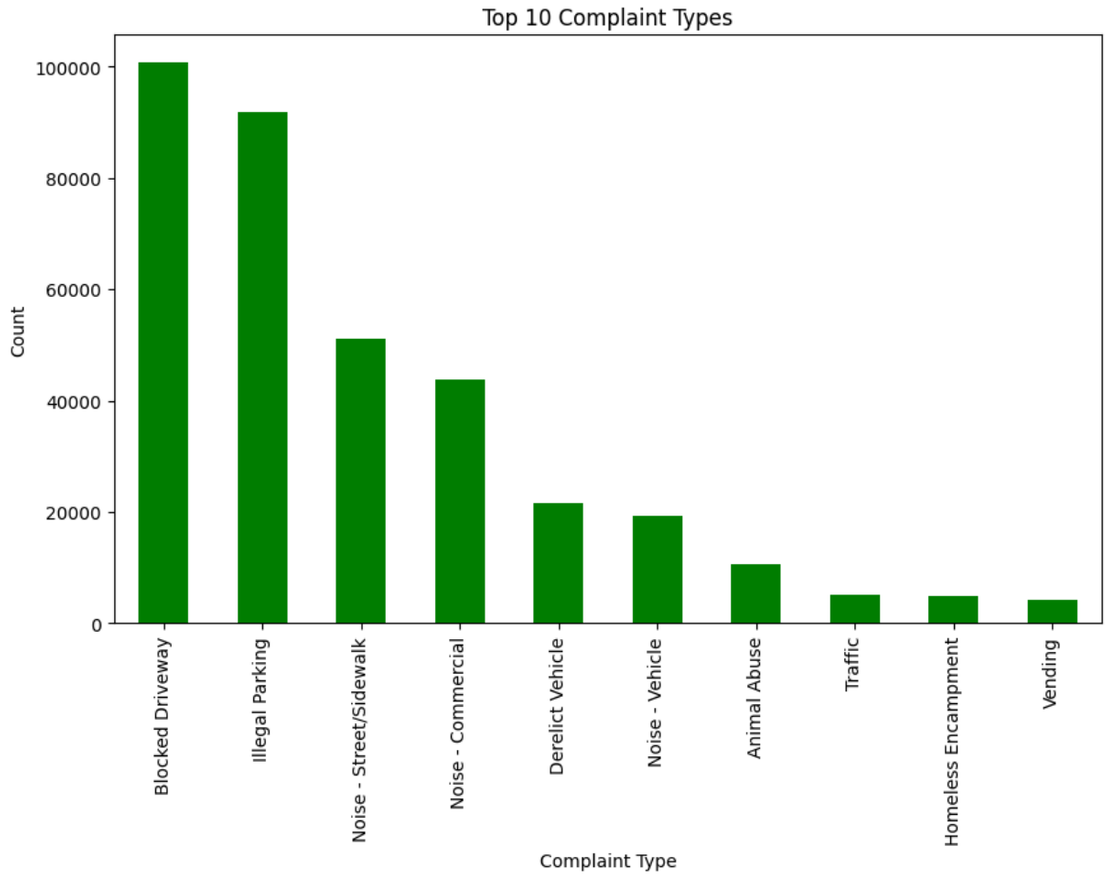
In [18]:
```
brooklyn = df[df['Borough'] == 'BROOKLYN']

plt.figure(figsize=(8,6))
plt.scatter(brooklyn['Longitude'], brooklyn['Latitude'], alpha=0.5, color = 'pink')
plt.title("Complaint Distribution in Brooklyn")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.show()
```



COMPLAINT ANALYSIS

In [19]:
```
plt.figure(figsize=(10,6))
df['Complaint Type'].value_counts().head(10).plot(kind='bar', color = 'green')
plt.title("Top 10 Complaint Types")
plt.xlabel("Complaint Type")
plt.ylabel("Count")
plt.show()
```



In [20]:
```
nyc = df[df['City'] == 'NEW YORK']

nyc['Complaint Type'].value_counts().head(10)
```

Out[20]:
```
Complaint Type
Noise - Street/Sidewalk    22245
Noise - Commercial         18686
Illegal Parking            14549
Noise - Vehicle             6294
Homeless Encampment         3060
Blocked Driveway            2705
Vending                     2638
Animal Abuse                1941
Traffic                     1769
Noise - Park                1243
Name: count, dtype: int64
```

In [21]:
```
top10 = df['Complaint Type'].value_counts().head(10)
top10
```

Out[21]:
```
Complaint Type
Blocked Driveway           100624
Illegal Parking             91716
Noise - Street/Sidewalk     51139
Noise - Commercial          43751
Derelict Vehicle            21518
Noise - Vehicle             19301
Animal Abuse                10530
Traffic                      5196
Homeless Encampment          4879
Vending                      4185
Name: count, dtype: int64
```

In [22]:
```
city_complaints = df.groupby(['City','Complaint Type']).size()
city_complaints
```

Out[22]:
```
City      Complaint Type
ARVERNE   Animal Abuse                46
          Blocked Driveway            50
          Derelict Vehicle            32
          Disorderly Youth             2
          Drinking                     1
                                     ...
Woodside  Blocked Driveway            27
          Derelict Vehicle             8
          Illegal Parking            124
          Noise - Commercial           2
          Noise - Street/Sidewalk      5
Length: 792, dtype: int64
```

Pivot Table

In [23]:
```
df_new = df.pivot_table(index='Complaint Type', columns='City', aggfunc='size', fil
df_new
```

Out[23]:

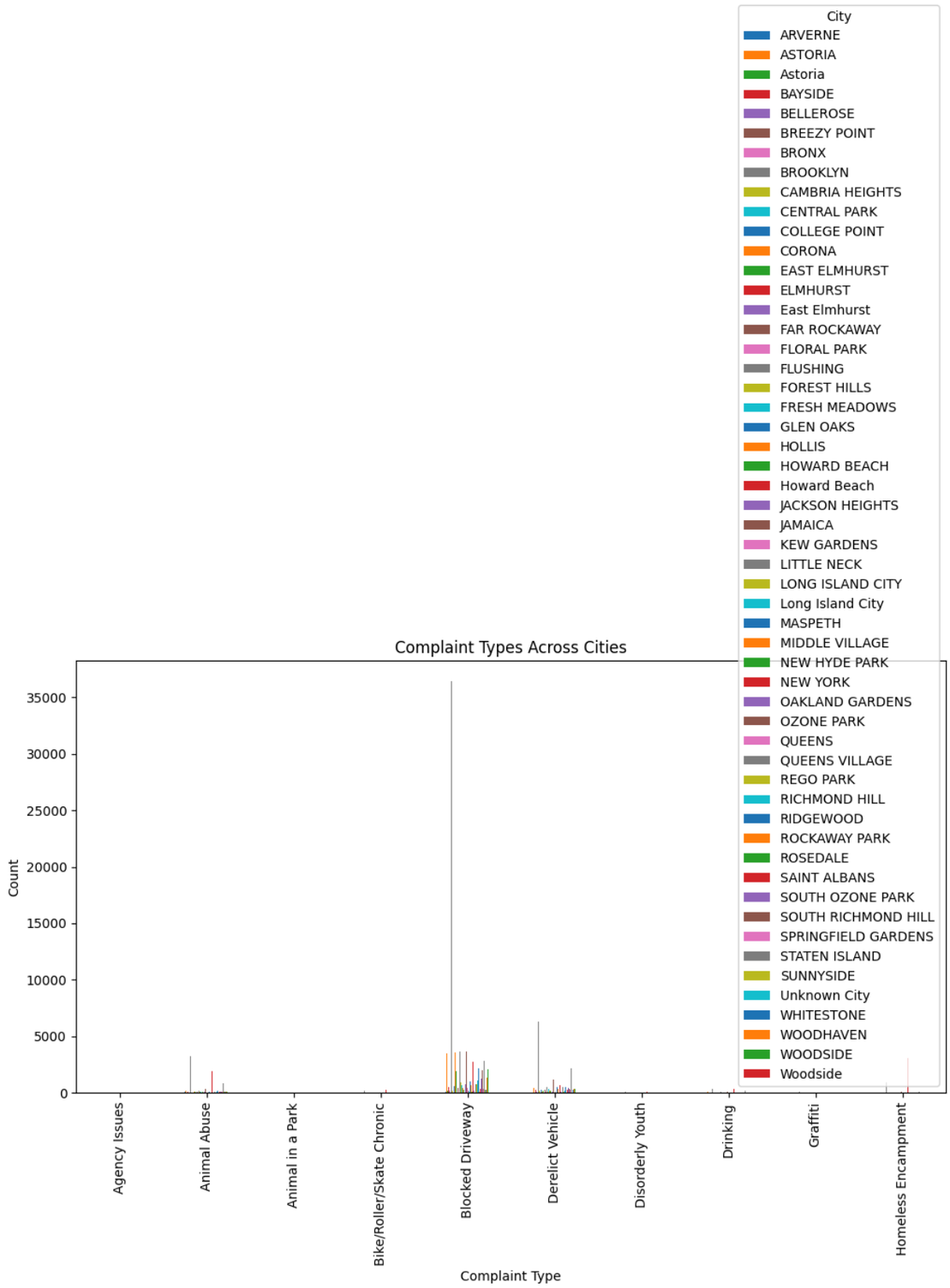
City	ARVERNE	ASTORIA	Astoria	BAYSIDE	BELLEROSE	BREEZY POINT	BRONX	BI
Complaint Type								
Agency Issues	0	0	0	0	0	0	0	
Animal Abuse	46	170	0	53	15	2	1971	
Animal in a Park	0	0	0	0	0	0	0	
Bike/Roller/Skate Chronic	0	16	0	0	1	0	22	
Blocked Driveway	50	3436	159	514	138	3	17062	
Derelict Vehicle	32	426	14	231	120	3	2402	
Disorderly Youth	2	5	0	2	2	0	66	
Drinking	1	43	0	1	1	1	206	
Graffiti	1	4	0	3	0	0	15	
Homeless Encampment	4	32	0	2	1	0	275	
Illegal Fireworks	0	4	0	0	1	0	24	
Illegal Parking	62	1340	277	638	132	16	9889	
Noise - Commercial	2	1653	310	47	38	4	2944	
Noise - House of Worship	14	21	0	3	1	0	90	
Noise - Park	2	64	0	4	1	0	548	
Noise - Street/Sidewalk	29	409	145	17	13	1	9144	
Noise - Vehicle	10	236	0	24	11	1	3556	
Panhandling	1	2	0	0	1	0	20	
Posting Advertisement	0	3	0	0	1	0	18	
Squeegee	0	0	0	0	0	0	0	
Traffic	1	60	0	9	9	0	427	
Urinating in Public	1	10	0	0	1	0	54	
Vending	1	57	0	2	0	0	433	

23 rows × 54 columns

VISUALIZATION

Complaint Types per City

In [24]:
```
df_new.head(10).plot(kind='bar', figsize=(12,6))
plt.title("Complaint Types Across Cities")
plt.ylabel("Count")
plt.show()
```



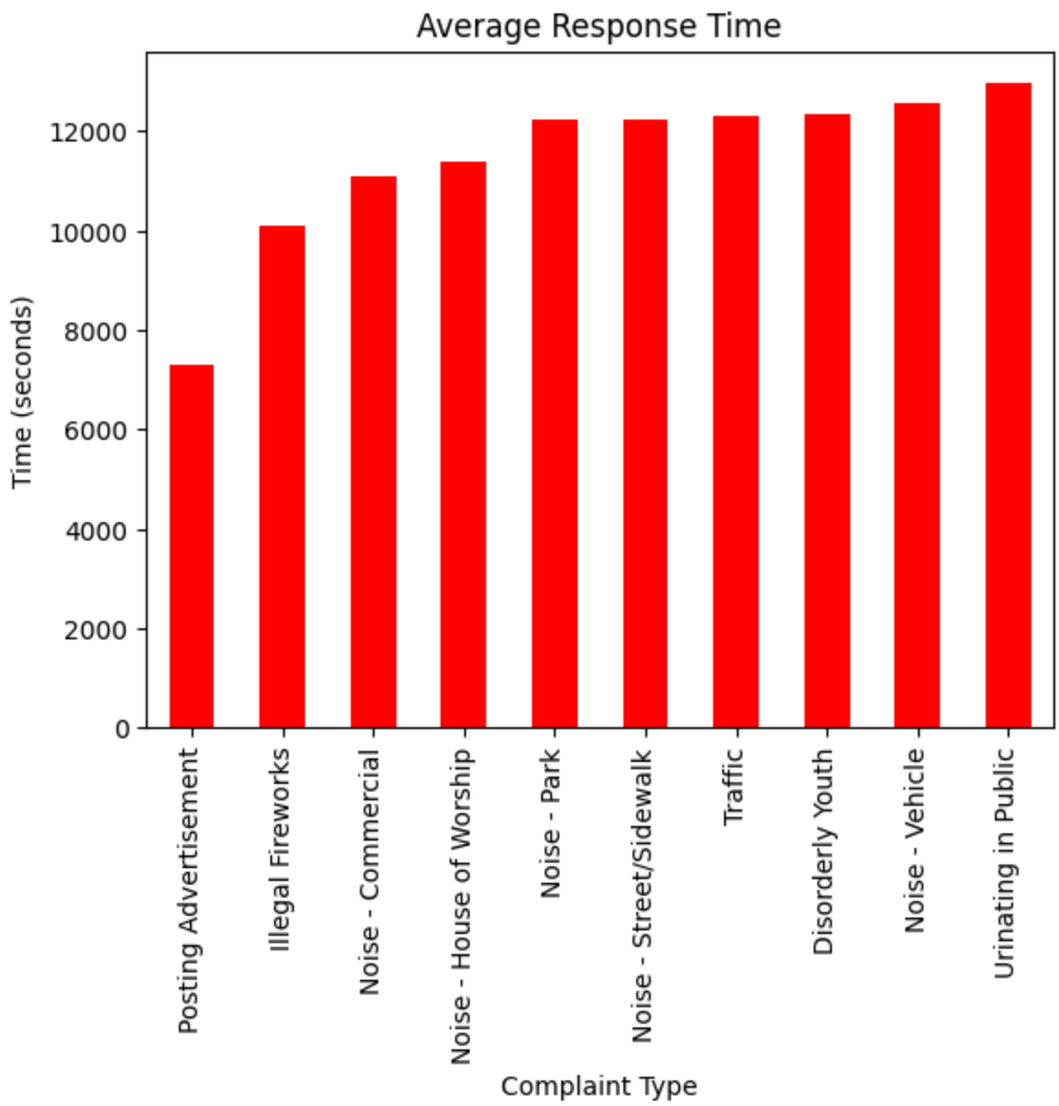
In [25]:
```
avg_time = df.groupby('Complaint Type')['Request_Closing_Time'].mean().sort_values(
avg_time.head(10)
```

Out[25]:
```
Complaint Type
Posting Advertisement        7286.256259
Illegal Fireworks           10113.482558
Noise - Commercial          11085.760531
Noise - House of Worship    11391.087079
Noise - Park                12226.055515
Noise - Street/Sidewalk     12231.295411
Traffic                     12309.120092
Disorderly Youth            12363.749206
Noise - Vehicle             12561.800010
Urinating in Public         12959.293292
Name: Request_Closing_Time, dtype: float64
```

Visualize Average Closing Time

In [26]:
```
avg_time.head(10).plot(kind='bar', color = 'red')
plt.title("Average Response Time")
```

```
plt.ylabel("Time (seconds)")
plt.show()
```



STATISTICAL TEST (Kruskal-Wallis)

In [27]:
```
groups = df.groupby('Complaint Type')['Request_Closing_Time'].apply(list)
```

Kruskal Test

In [28]:
```
from scipy.stats import kruskal

kruskal(*groups[:5])
```

Out[28]:
```
KruskalResult(statistic=np.float64(65.47298435906306), pvalue=np.float64(2.0456061
161139882e-13))
```

Interpretation

Based on the Kruskal-Wallis test, the p-value is less than 0.05. Therefore, we reject the null hypothesis (H0). This indicates that there is a significant difference in response times across different complaint types.

OBSERVATIONS

1. The dataset contains multiple types of complaints across different cities.

2. The most frequent complaints are related to noise, illegal parking, and blocked driveways.
3. New York City has the highest number of complaints compared to other cities.
4. The average response time varies depending on the type of complaint.
5. Some complaints are resolved quickly, while others take longer time.
6. Based on the Kruskal-Wallis test, there is a significant difference in response times across complaint types (if p-value < 0.05).

In [ ]:

```

```